

POST-VACCINAL ENCEPHALITIS.

REPORT BY THE LEAGUE OF NATIONS COMMISSION.

THE Commission on Small-pox and Vaccination of the Health Organization of the League of Nations has issued a memorandum reporting the findings of two of its sub-committees, one on post-vaccinal encephalitis, and the other on the preparation and preservation of vaccine lymph. The former report derives much of its information from English sources, and quotes extensively from the Rolleston Committee's report that was summarized in the *British Medical Journal* of August 11th, 1928 (p. 266).

The Commission emphasizes the fact that post-vaccinal does not necessarily imply propter-vaccinal; that in view of the millions of vaccinations that are carried out in any given year there must always be present in the community very large numbers of people subject to the ordinary chances of life, and *incidentally* under the influence of the vaccinia virus. Such people are no more immune from disease of the brain and nervous system than anyone else, and there is, therefore, no reason why the coincidence of vaccinia with a true encephalitis should not be expected to occur. Encephalitis, moreover, occurs as a complication of numerous infections—for example, measles and various forms of toxæmia—and in so far as vaccinia is an infection it might be considered legitimate to regard the cases under consideration merely as instances of ordinary post-infective encephalitis. The Commission, however, is of opinion that there are at present practical reasons for considering post-vaccinal encephalitis as a separate pathological and clinical entity. Particularly in Holland and in England have the cases been numerous enough to merit collective consideration.

The Comparative Incidence of Cases.

The risk of post-vaccinal encephalitis may easily be exaggerated; indeed, in Holland it has been considered sufficiently grave to cause temporary suspension of the laws relating to the vaccination of children, and in England it has received the attention of two separate Ministry of Health Committees—the Andrewes Committee of 1924, and the Rolleston Committee of 1926. In actual fact, during the whole period 1923–27, only 139 cases of disease of the central nervous system following vaccination were reported in Holland, 41 being fatal. In England and Wales 62 cases, with 36 deaths, occurred between November, 1922, and November, 1923, and a further 25 cases, with 12 deaths, between January, 1926, and September, 1927—the periods reported on by the two committees.

The age distribution in the incidence of post-vaccinal encephalitis is of particular interest. The bulk of the cases occur in children between the ages of 3 and 13—infants and adults being almost wholly immune. Moreover, the disease follows the revaccinations of older children and of adults with the utmost rarity, a fact which, taken in conjunction with the almost complete immunity of the first two years, provides ample reason for emphasizing the value of undertaking prophylactic vaccination against small-pox as early as possible in life.

Outstanding Clinical Features.

In nearly every instance the complication has occurred in cases of vaccinia of normal appearance, apparently running a normal course. No single method of vaccination has been specially implicated, nor have any particular strains of lymph or methods of preparation been concerned. A noteworthy fact is that both in England and Wales and in Holland it has been especially in the less populous rural areas that vaccination has been followed by encephalitic complications.

The correspondence observed in the early Dutch cases between post-vaccinal encephalitis and the local distribution of true encephalitis lethargica has not been observed in England or maintained in Holland; indeed, the clinical history and pathological findings of cases of post-vaccinal encephalitis justify the conclusion that one is dealing with two separate diseases. Interesting clinical features of post-vaccinal encephalitis are the incubation period—which is between the ninth and thirteenth day following vaccination, with marked preference shown for the eleventh day—and the sudden onset with four cardinal symptoms

of headache, vomiting, drowsiness, and pyrexia. The extremes met with are mild cases which apparently recover completely in a week (though in a few of these mental deterioration and residual paralysis have been reported) and severe cases which develop extensive spastic paralysis and pass rapidly into coma and death. Between these extremes are met cases of every grade of severity. The cerebro-spinal fluid is commonly under pressure but is sterile. Pathological examination of fatal cases shows the presence of diffuse encephalo-myelitis with perivascular infiltration and oedema. In some cases the histological changes resemble those of acute disseminated sclerosis; and in all cases they differ essentially from those met with in true epidemic encephalitis.

The general conclusion of the Commission is that the virus of vaccinia cannot of itself be considered responsible for encephalitis. It considers rather that some unknown factor—perhaps a filter-passing virus or a latent virus—determines the occurrence of the complication in question.

PREPARATION AND PRESERVATION OF LYMPH.

The sub-commission on the preparation and preservation of lymph dealt with the factors which may favourably or unfavourably affect its value as a vaccine. It concluded that only the treatment to which the lymph seed is subjected determines the quality of the lymph, the origin of the strain used being of no significance in this connexion.

The virulence of the vaccine virus is sometimes maintained when it is carried in unbroken series from calf to calf; more usually it is found necessary to interpolate some other passage in the series, commonly the rabbit, or to secure a retro-vaccine, commonly a lymph obtained from man. These methods appear to be unobjectionable. On the other hand, neuro-lapine obtained by intra-cerebral injections into rabbits, and testicular lapine, sometimes display properties that differ from ordinary vaccine, and conclusions as to their effects on man cannot yet be formulated.

Veterinary Control.

The Commission, while not dealing with the technique of preparing lymph ready for issue, emphasizes the vital need for veterinary control of the animals used, such control involving examination of the living animal before inoculation and the slaughter and *post-mortem* examination of the animal as soon as the vaccine has been obtained. The lymph should be used only when the results of these examinations are unexceptionable, and, in particular, when there are no signs of tuberculous lesions.

Preparation and Employment of Lymph.

Questions relating to the employment of diluted vaccine have been specially considered. Dilution of lymph does not necessarily imply loss of virulence, it being better to employ a strain of more than minimal standard, diluted, than one of weak virulence undiluted. But investigations still need to be made into the durability of dilution, the nature of the diluent, the degree of dilution, and the actual clinical results obtained from diluted lymph.

The Commission is strongly of opinion that the medical officers of vaccine institutes should themselves undertake human vaccination in order to study the potency of the lymph they make. The bare report "successful or unsuccessful" is of no value when the vaccine reaction is not described to them in detail. In primary vaccination the percentage of *insertion successes* should be noted; in revaccinations the reaction should be described in official records as one of three classes:

1. Reaction of the primary vaccination type (Jenner's pustule).
2. Accelerated reaction (vesicular reaction); "modified vaccine."
3. Allergic or early reaction (papular reaction).

The Commission recommends that the technique and methods of vaccination permitted and recommended in different countries should be carefully defined, the results duly recorded, and then judged in respect of actual immunity afforded on exposure to small-pox or on revaccination. In this way a satisfactory technique for universal adoption may ultimately be evolved. It concludes by reasserting its conviction that vaccination remains the most powerful weapon against small-pox we possess, and strongly advocates primary vaccination in early infancy rather than at the later ages of childhood.